Concept Note: Development of an Autonomous Multi-Agent Framework for OpenFOAM Case Generation

1. Background

Computational Fluid Dynamics (CFD) is a critical tool in mechanical engineering for analyzing complex fluid flows, heat transfer, and fluid-structure interactions. Our research group heavily relies on **OpenFOAM**, a highly robust, open-source C++ CFD framework. While powerful, OpenFOAM lacks a traditional Graphical User Interface (GUI). Instead, simulations are driven by a strictly formatted, heavily interlinked directory structure containing text-based dictionary files (defining meshing, boundary conditions, fluid properties, and numerical schemes). The learning curve for manual case setup is notoriously steep and prone to syntax errors.

Recent advancements in local Large Language Models (LLMs) and multi-agent orchestrations (e.g., LangChain, AutoGen, CrewAI) offer a novel solution. By treating the OpenFOAM directory structure as a programmatic output, we can use agentic workflows to translate plain English descriptions of a physical problem directly into a ready-to-run simulation directory. This project aims to bridge the gap between natural language prompts and strict OpenFOAM syntax, utilizing a secure, locally hosted AI environment.

2. Project Objectives

The primary aim of this two-month internship is to architect, test, and document a multi-agent AI pipeline capable of autonomous OpenFOAM case generation. The specific objectives are to:

- **Bridge the Domain Gap**
  - Understand the basic data architecture, file dependencies, and syntax rules of OpenFOAM (specifically the OpenFOAM.org foundation version).
- **Establish Secure Infrastructure**
  - Deploy and configure a local LLM environment on an isolated Ubuntu Virtual Machine via standard APIs.
- **Architect an Agentic Workflow**

- Develop a multi-agent Python framework where specialized agents collaborate to generate meshes, define boundary conditions, and configure solver settings based on natural language inputs.

- **Package and Standardize**
- Deliver a modular, well-documented codebase and technical report to facilitate future development and seamless integration into the research group's daily workflow.

3. Definitive Tasks and Timeline (8-Week Program)

Phase I: Infrastructure Deployment & Domain Onboarding (Weeks 1–2)

- Complete a rapid onboarding regarding OpenFOAM's case directory structure (0/, constant/, system/) and standard meshing procedures (e.g., blockMesh).
- Provision and configure the isolated Ubuntu VM.
- Deploy the required local, open-source LLM resources (e.g., via Ollama or vLLM) and establish a stable connection to the local model API.

Phase II: Agentic Framework Initialization (Weeks 3–4)

- Set up the core Python environment and install the chosen agentic framework libraries.
- Design the multi-agent topology (e.g., assigning a "Meshing Agent," a "Physics Agent," and a "Syntax Validation Agent").
- Write the initial system prompts and provide the agents with the necessary context and open-source tools to read/write system files.

Phase III: Workflow Refinement & Iterative Testing (Weeks 5–6)

- Feed detailed natural language simulation descriptions into the workflow (e.g., "Set up a 2D transient simulation of air flowing over a cylinder at 2 meters per second").
- Iteratively tweak agent prompts, memory handling, and tool-use to ensure the generated text dictionaries adhere strictly to OpenFOAM C++ syntax.
- Implement a feedback loop where the agent attempts a "dry run" (e.g., running the blockMesh command) and self-corrects based on the terminal error output.

Phase IV: Documentation, Packaging, and Handoff (Weeks 7–8)

- Finalize the multi-agent codebase, ensuring it consistently produces simple, runnable case structures.
- Package the environment and models appropriately (using Docker, Git, or virtual environment freezing) for scalable deployment.

- Draft a comprehensive technical report detailing the chosen tools, the agent architecture, failure modes, and recommendations for future scalability.

4. Key Performance Indicators (KPIs)

- To ensure clear expectations and objective evaluation of the candidate's progress, the following KPIs will be tracked:

- **Infrastructure Milestone:**
    - Successful deployment of the Ubuntu VM and a responsive local LLM API by the end of Week 2.
- **Pipeline Alpha Milestone**
    - The multi-agent workflow successfully takes a text prompt and generates the basic three-folder OpenFOAM directory structure by the end of Week 4.
- **Syntax Validation**
    - By Week 6, the agent-generated case structure must successfully pass OpenFOAM's native syntax checks (e.g., executing blockMesh without fatal C++ dictionary formatting errors) for a baseline test case.
- **Handoff Quality**
    - Delivery of a fully reproducible codebase (with a clear README and dependency list) alongside a structured technical report by the conclusion of Week 8.